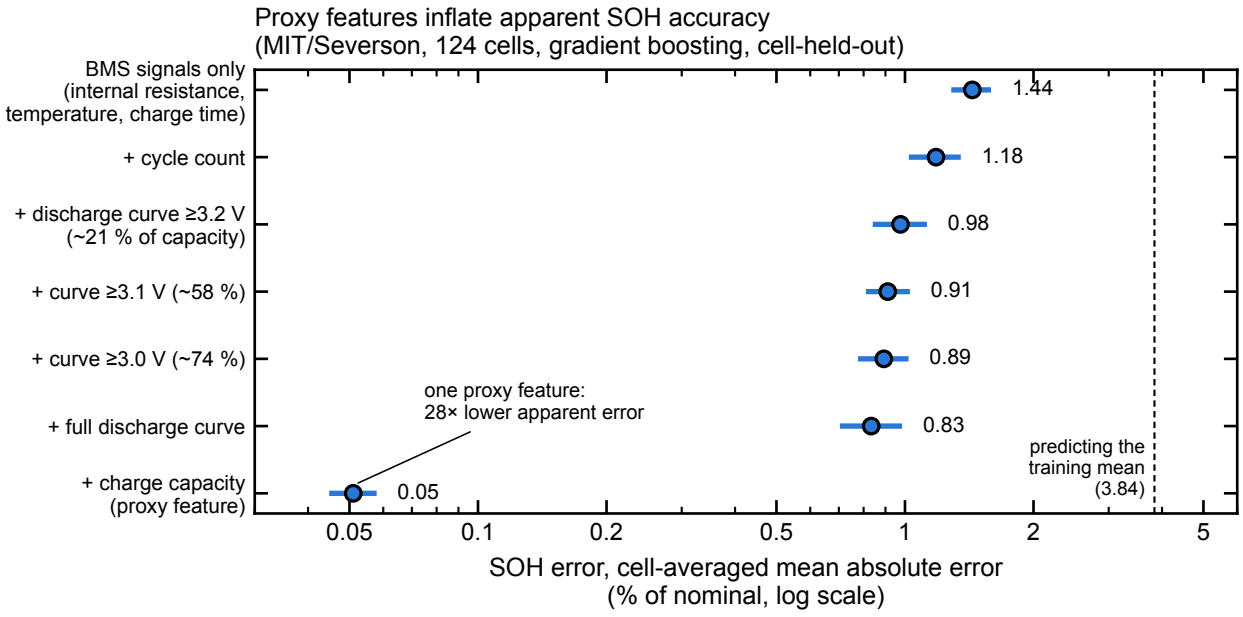
Proxy features inflate apparent SOH accuracy
(MIT/Severson, 124 cells, gradient boosting, cell-held-out)
BMS signals only (internal resistance, temperature, charge time)
+ cycle count
+ discharge curve ≥3.2 V (~21 % of capacity)
+ curve ≥3.1 V (~58 %)
+ curve ≥3.0 V (~74 %)
+ full discharge curve
+ charge capacity (proxy feature)
1.44
1.18
0.98
0.91
0.89
0.83
0.05
one proxy feature:
28× lower apparent error
predicting the training mean (3.84)
0.05
0.1
0.2
0.5
1
2
5
SOH error, cell-averaged mean absolute error
(% of nominal, log scale)